

MARK ABRAHAM MAGUMBA

1. **Name:** MARK ABRAHAM MAGUMBA
2. **Profession:** Researcher/ ICT expert/ Data Scientist/ Educator/ Health informatician
3. **Education:**
 - **PhD in Information Systems, Makerere University (Uganda)**, Areas: Knowledge Engineering, Machine Learning and Natural Language Processing in Health Informatics, titled Ontology Based Language Modelling for Generalized Automated Disease Incidence Detection in The Social Web, (Graduated May 2021)
 - **MSc IT management, University of Sunderland (United Kingdom)**, December 2009
 - **BSc Information Technology, Makerere University (Uganda)**, September 2007
4. **Nationality:** UGANDAN
5. **Employment Record**

September 2023 To date

Employer: Technological University Dublin, Republic of Ireland

Position Held: Postdoctoral researcher

Summary:

- Conduct research on scalable and energy-efficient AI systems using neural network compression techniques including quantization, pruning, federated learning, collaborative inference, and neural network re-architecting.
- Design and implement machine learning workflows and experimental pipelines for edge AI applications and resource-constrained deployment scenarios.
- Investigate techniques for improving inference efficiency, model scalability, and energy consumption of deep learning systems.
- Collaborate with multidisciplinary research teams and industry stakeholders on AI-focused innovation projects.
- Deliver technical feasibility studies and AI consultancy through industry engagement initiatives.
- Manage Linux-based development infrastructure including software deployment, system administration, and security hardening.
- Publish research findings in peer-reviewed journals and international conferences.

August 2012 – December 2025

Employer: Makerere University College of Computing and Information Sciences, Uganda

Position Held: Lecturer / Assistant Lecturer

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- Collaborate with multidisciplinary research teams and industry stakeholders on AI-focused innovation projects.
- Deliver technical feasibility studies and AI consultancy through industry engagement initiatives.
- Manage Linux-based development infrastructure including software deployment, system administration, and security hardening.
- Publish research findings in peer-reviewed journals and international conferences.

January 2013 – January 2014

Employer: National Forum for People Living with HIV/AIDS Networks in Uganda

Position Held: ICT Consultant

Summary:

- Conducted systems analysis and requirements gathering.
- Designed and implemented IT solutions to improve operational efficiency
- Provided stakeholder-facing technical consultancy and deployment support.

June 2010 – April 2011

Employer: Movit Products Limited

Position Held: ICT Manager

Summary:

- Managed enterprise IT infrastructure, systems administration, security, and software support.
- Coordinated development of ERP and biometric attendance solutions integrated with HR workflows.
- Implemented IT policies, backup strategies, and operational security controls.